

Parabola

Question1

If $x - y - 3 = 0$ is a normal drawn through the point $(5, 2)$ to the parabola $y^2 = 4x$, then the slope of the other normal that can be drawn through the same point to the parabola $y^2 = 4x$ is

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Options:

A.

0

B.

-1

C.

2

D.

-2

Answer: D

Solution:

$$\because y^2 = 4x$$

$$\text{So, } a = 1$$

$\therefore$ The equation of the normal

$$y = mx - 2am - am^3$$

$$\Rightarrow y = mx - 2m - m^3$$

which passes through $(5, 2)$

$$\therefore 2 = 5m - 2m - m^3$$

$$\Rightarrow m^3 - 3m + 2 = 0$$

and the given normal is

$$y = x - 3$$

$\therefore m = 1$, put the value

$$(1)^3 - 3 \times 1 + 2 = 0$$

Thus, $m = 1$ is a root.

$$\therefore m^3 - 3m + 2 = (m - 1)^2(m + 2) = 0$$

So, there are two normals with slopes 1 and -2

$\therefore$ The given normal has slope 1

So, the other normal has slope $= -2$

Question2

If the normal chord drawn at the point $\left(\frac{15}{2}, \frac{15}{\sqrt{2}}\right)$ to the parabola $y^2 = 15x$ subtends an angle θ at the vertex of the parabola, then $\sin \frac{\theta}{3} + \cos \frac{2\theta}{3} - \sec \frac{4\theta}{3} =$

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Options:

A.

0

B.

3

C.

1

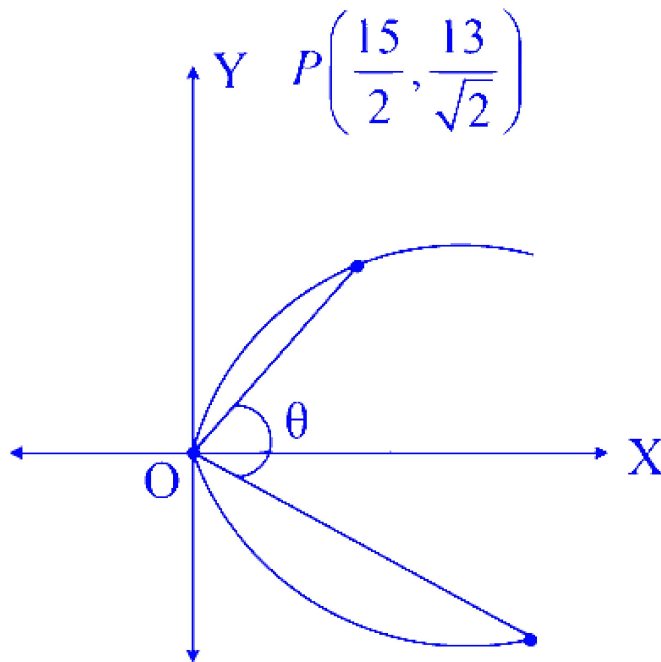
D.

Answer: B

Solution:

$$y^2 = 15x$$

Equation of normal at $\left(\frac{15}{2}, \frac{15}{\sqrt{2}}\right)$



$$2y \frac{dy}{dx} = 15$$

$$\Rightarrow \frac{dy}{dx} = \frac{15}{2y}$$

$$\Rightarrow m_N = \frac{-1}{\frac{dy}{dx}} = \frac{-2y}{15} \quad \left(\because y = \frac{15}{\sqrt{2}} \right)$$

$$\Rightarrow m_N = \frac{-2 \times 15/\sqrt{2}}{15} = -\sqrt{2}$$

Equation of normal

$$y - \frac{15}{\sqrt{2}} = -\sqrt{2} \left(x - \frac{15}{2} \right)$$

$$\Rightarrow \sqrt{2}y - 15 = -2 \left(x - \frac{15}{2} \right)$$

$$\Rightarrow \sqrt{2}y - 15 = -2x + 15$$

$$\Rightarrow 2x + \sqrt{2}y = 30$$

$$\Rightarrow \frac{2x + \sqrt{2}y}{30} = 1$$

Homogenisation of $y^2 = 15x$ w.r.t L

$$y^2 = 15x \left(\frac{2x + \sqrt{2}y}{30} \right)$$

$$\Rightarrow 2y^2 = 2x^2 + 2\sqrt{2}xy$$

$$\Rightarrow 2x^2 + 2\sqrt{2}xy - 2y^2 = 0$$

$$\therefore a + b = 0$$

$$\therefore \theta = \pi/2$$

$$\begin{aligned} \text{Now, } \sin \frac{\theta}{3} + \cos \frac{2\theta}{3} - \sec \frac{4\theta}{3} \\ = \sin \frac{\pi}{6} + \cos \frac{\pi}{3} - \sec \left(\frac{2\pi}{3} \right) \\ = 1/2 + 1/2 + 2 = 3 \end{aligned}$$

Question3

Tangents are drawn at three points $P(t_1)$, $Q(t_2)$, $R(t_3)$ on the parabola $y^2 = x$. Let these tangents intersect each other at the points L, M, N . If $t_1 = 2, t_2 = -4, t_3 = 6$, then the area of the $\triangle LMN$ is

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Options:

A.

24

B.

18.5

C.

7.5

D.

12

Answer: C

Solution:

Given, parabola is $y^2 = x$ and $t_1 = 2t_2 = -4$ and $t_3 = 6$.

So, intersection of tangents at t_1 and t_2 , then tangents are $2y = x + 1$ and $-4y = x + 4$

Solving these two equations, we get $y = \left(-\frac{1}{2}\right)$ and $x = -2$

So, point $L = \left(-2, -\frac{1}{2}\right)$

Now, intersection of tangents at $t_2 = -4$ and $t_3 = 6$, then tangents are

$$-4y = x + 4, 6y = x + 9$$

Solving these two,

$$\text{we get } x = -6, y = \frac{1}{2}$$

So, point $M = \left(-6, \frac{1}{2}\right)$

Now, intersection of tangents at $t_1 = 2$ and $t_3 = 6$, so tangents are $2y = x + 1$ and $6y = x + 9$

Solving these two equations, we get $x = 3, y = 2$

So, point $N = (3, 2)$

Area of triangle with vertices $L\left(-2, -\frac{1}{2}\right)$, $M\left(-6, \frac{1}{2}\right)$ and $N(3, 2)$ is

$$\begin{aligned}\text{Area} &= \left| \frac{1}{2} \begin{vmatrix} -2 & -\frac{1}{2} & 1 \\ -6 & \frac{1}{2} & 1 \\ 3 & 2 & 1 \end{vmatrix} \right| \\ &= \left| \frac{-15}{2} \right| = 7.5\end{aligned}$$

Question4

If the tangents of the parabola $y^2 = 8x$ passing through the point $P(1, 3)$ touches the parabola at A and B , then the area (in sq. units) of $\triangle PAB$ is

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Options:

A.

1

B.

$$\frac{3}{4}$$

C.

$$\frac{1}{2}$$

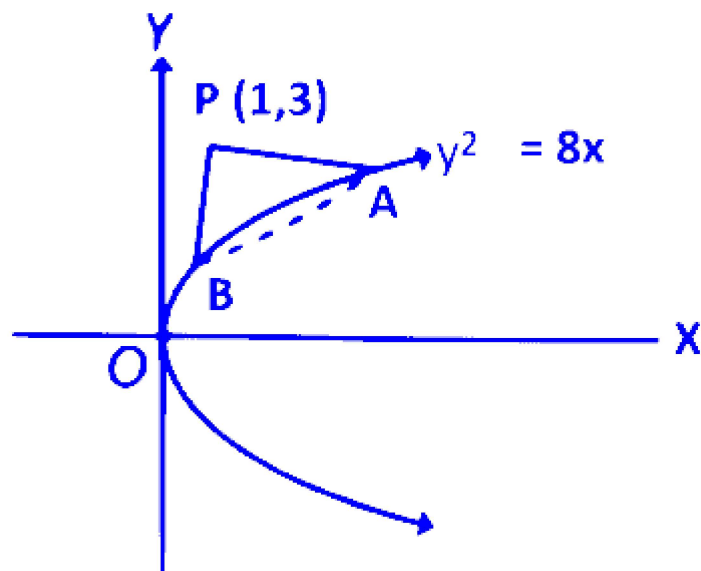
D.

$$\frac{1}{4}$$

Answer: D

Solution:

$$y^2 = 8x, a = 2$$



Equation of tangent

$$y = mx + \frac{2}{m}$$

at (1,3)

$$3 = m + \frac{2}{m} \Rightarrow m^2 - 3m + 2 = 0$$

$$\Rightarrow m = 1, 2$$

So, equation of tangent (put point (1, 3) on $y = mc + c$)

$$y = x + c, y = 2x + c$$

$$\Rightarrow y = x + 2y = 2x + 1$$

to find coordinates of A and B, put the line on parabola.

$$(x + 2)^2 = 8x \Rightarrow x^2 - 4x + 4 = 0$$

$$\Rightarrow x = 2, -2$$

$$\text{at } x = 2, y = \sqrt{8 \times 2} = 4$$

So, $A = (2, 4)$

and $(2x + 1)^2 = 8x$

$$\Rightarrow 4x^2 - 4x + 1 = 0$$

$$\Rightarrow x = \frac{1}{2}, -\frac{1}{2}$$

at $x = \frac{1}{2}, y = 2$

So, $B = \left(\frac{1}{2}, 2\right)$

Now, area of $\triangle PAB$

$$\begin{aligned} A &= \frac{1}{2} \begin{vmatrix} 1 & 3 & 1 \\ 2 & 4 & 1 \\ \frac{1}{2} & 2 & 1 \end{vmatrix} \\ &= \left| \frac{1}{2} \left(1(4 - 2) - 3 \left(2 - \frac{1}{2} \right) + 1(4 - 2) \right) \right| \\ &= \left| \frac{1}{2} \left(2 - \frac{9}{2} + 2 \right) \right| = \left| \frac{1}{2} \left(4 - \frac{9}{2} \right) \right| \\ &= \left| \frac{1}{2} \left(-\frac{1}{2} \right) \right| = \frac{1}{4} \end{aligned}$$

Question5

The lengths of the two focal chords of the parabola $y^2 = 16x$ is 25 units each. If these two chords cut the parabola at A, B, C and D , then the area (in sq. units) of the quadrilateral formed by A, B, C and D is

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Options:

A.

$$\frac{625}{2}$$

B.

180

C.

D.

300

Answer: D**Solution:**

Given equation of parabola is

$$y^2 = 16x \quad \dots (i)$$

Comparing it with $y^2 = 4ax$ we get, $a = 4$

Let coordinates of end points of focal chord of a parabola (i) is, $(at^2, 2at)$ and $(\frac{a}{t^2}, \frac{-2a}{t})$

$$\begin{aligned} &= \text{its length} = \sqrt{\left(at^2 - \frac{a}{t^2}\right)^2 + \left(2at + \frac{2a}{t}\right)^2} \text{ (given)} \\ &= 25 \end{aligned}$$

$$\Rightarrow \sqrt{\left(4\left(t^2 - \frac{1}{t^2}\right)\right)^2 + \left(8\left(t + \frac{1}{t}\right)\right)^2} = 25$$

(putting $a = 4$)

$$\Rightarrow 16\left(t^2 - \frac{1}{t^2}\right)^2 + 64\left(t + \frac{1}{t}\right)^2 = 625$$

$$\Rightarrow 16\left(t^4 + \frac{1}{t^4} - 2\right) + 64\left(t^2 + \frac{1}{t^2} + 2\right) = 625 \quad \dots (ii)$$

Let $t + \frac{1}{t} = u$ so that $\left(t + \frac{1}{t}\right)^2 = u^2$

$$\Rightarrow t^2 + \frac{1}{t^2} + 2 = u^2$$

$$\Rightarrow t^2 + \frac{1}{t^2} = u^2 - 2$$

$$\Rightarrow \left(t^2 + \frac{1}{t^2}\right)^2 = (u^2 - 2)^2$$

$$\Rightarrow t^4 + \frac{1}{t^4} + 2 = u^4 + 4 - 2u^2$$

Putting these values in Eq. (ii) and solving then we get,

$$t = 2, t = \frac{1}{2}, t = -2, t = \frac{-1}{2}$$

So, Points are $A(16, 16)$, $B(1, 4)$, $C(1, -4)$, $D(16, -16)$

$\therefore$ Area of quadrilateral $ABCD =$

$$\begin{aligned}
& \frac{1}{2} \left| x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1 - (y_1x_2 + y_2x_3 + y_3x_4 + y_4x_1) \right| \\
&= \frac{1}{2} \left| 16 \times 4 + 1 \times (-4) + 1 \times (-16) + 16 \times 16 - (16 \times 1 + 4 \times 1 + (-4) \times 16 + (-16)(16)) \right| \\
&= 300 \text{ sq. units}
\end{aligned}$$

Question6

If the perpendicular distance from the focus of a parabola $y^2 = 4ax$ to its directrix is $\frac{3}{2}$, then the equation of the normal drawn at $(4a, -4a)$ is

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Options:

A.

$$2x + y = 3$$

B.

$$2x - y = 9$$

C.

$$x - 2y = 9$$

D.

$$x + 2y + 3 = 0$$

Answer: B

Solution:

$$y^2 = 4ax, \text{ focus} = (a, 0)$$

Equation of directrix is $x = -a$.

The perpendicular distance from the focus $(a, 0)$ to the directrix $x + a = 0$

$$\Rightarrow \frac{|a+a|}{\sqrt{1+0}} = |2a| = |2a|$$

$$\Rightarrow |2a| = \frac{3}{2}$$

$$\Rightarrow 2a = \frac{3}{2} \text{ or } 2a = \frac{-3}{2}$$

$$\Rightarrow 2a = \frac{3}{2} \Rightarrow a = \frac{3}{4}$$

$$\Rightarrow y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

Point $(4a, -4a)$

$$x_1 = 4a, y_1 = -4a$$

$$\Rightarrow y - (-4a) = -\frac{(-4a)}{2a}(x - 4a)$$

$$\Rightarrow y + 4a = -(-2)(x - 4a)$$

$$\Rightarrow y + 4a = 2(x - 4a)$$

$$\Rightarrow y + 4a = 2x - 8a$$

$$\Rightarrow 2x - y - 12a = 0$$

Substituting $a = \frac{3}{4}$

$$\Rightarrow 2x - y - 12\left(\frac{3}{4}\right) = 0$$

$$\Rightarrow 2x - y - 9 = 0$$

$$2x - y = 9$$

Question7

PQ is a focal chord of the parabola $y^2 = 4x$ with focus S . If $P = (4, 4)$, then $SQ =$

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Options:

A.

2

B.

$\frac{5}{4}$

C.

5

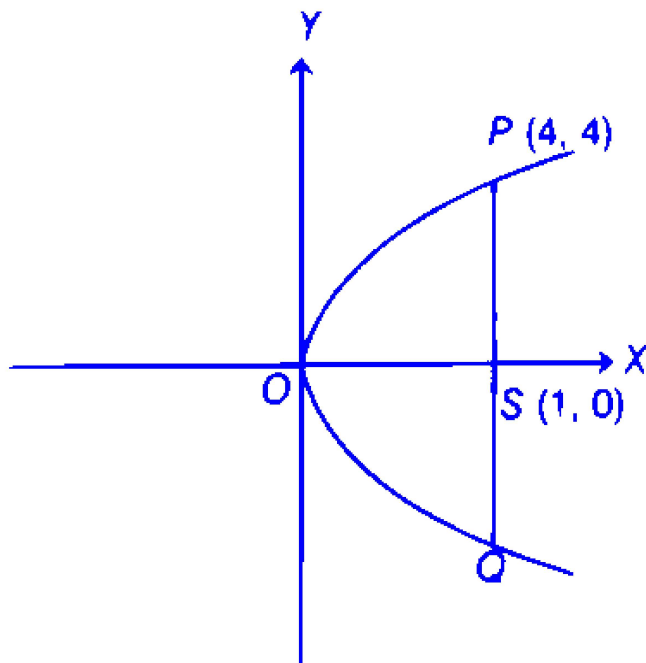
D.

$$\frac{3}{2}$$

Answer: B

Solution:

Given parabola $y^2 = 4x$



$$S \equiv (1, 0), \quad a = 1$$

Other end of focal chord Q

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a} \Rightarrow$$

$$\Rightarrow \frac{1}{5} + \frac{1}{SQ} = \frac{4}{5} \Rightarrow SQ = \frac{5}{4}$$

Question8

The angle between the tangents drawn from the point $(1, 4)$ to the parabola $y^2 = 4x$ is

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Options:

A.

$$\frac{\pi}{6}$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{3}$$

D.

$$\frac{\pi}{2}$$

Answer: C

Solution:

$$y^2 = 4x$$

Equation of tangent to the parabola

$$y^2 = 4ax \text{ is } y = mx + \frac{a}{m}$$

$\therefore$ Equation of tangent is

$$y = mx + \frac{1}{m}$$

Given, tangents passes through $(1, 4)$

$$4 = m + \frac{1}{m}$$

$$\Rightarrow m^2 - 4m + 1 = 0$$

$$\Rightarrow m_1 + m_2 = 4, \quad m_1 m_2 = 1$$

$$\therefore \tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}$$

$$= \frac{\sqrt{16 - 4}}{1 + 1} = \frac{\sqrt{12}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$\therefore \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$

Question9

If L is the normal drawn to the parabola $y^2 = 8x$ at the point $t = \frac{1}{\sqrt{2}}$, then the foot of the perpendicular drawn from the focus of the parabola on to the normal L is

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Options:

A.

$$(3, 2)$$

B.

$$(5, \sqrt{2})$$

C.

$$(0, \sqrt{2})$$

D.

$$(3, \sqrt{2})$$

Answer: D

Solution:

We have parabola, $y^2 = 8x$

Equation of normal at $(at^2, 2at)$ is

$$y + xt = 2at + at^3$$

$$\text{But } t = \frac{1}{\sqrt{2}}$$

$$y + x \frac{1}{\sqrt{2}} = 4 \frac{1}{\sqrt{2}} + \frac{2}{2\sqrt{2}}$$

$$\sqrt{2}y + x = 5$$

It is equation of L .

Focus of parabola $y^2 = 8x$ is $(2, 0)$

Foot of perpendicular from focus to the line L is

$$\frac{x-2}{1} = \frac{y-0}{\sqrt{2}} = -\left(\frac{2-5}{3}\right) = 1$$
$$x = 3 \text{ and } y = \sqrt{2}$$

$\therefore$ Foot of perpendicular = $(3, \sqrt{2})$.

Question10

If P is a point which divides the line segment joining the focus of the parabola $y^2 = 12x$ and a point on the parabola in the ratio 1 : 2. Then, the locus of p is

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Options:

A. $y^2 = 2(x - 2)$

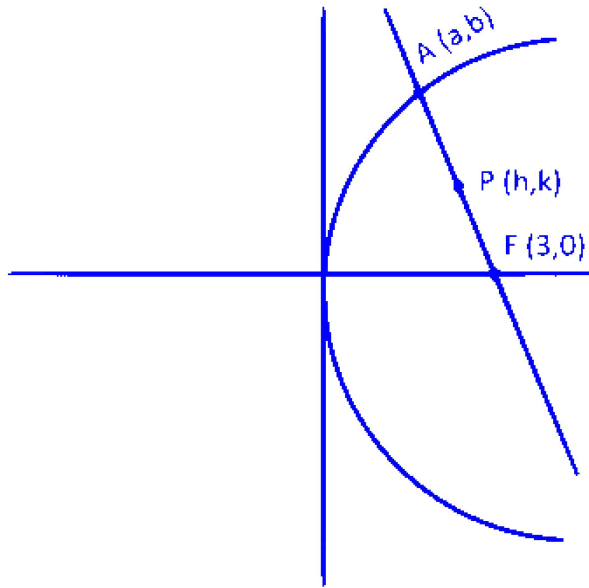
B. $y^2 = 4x$

C. $y^2 = 4(x - 2)$

D. $y^2 = 9(x - 3)$

Answer: C

Solution:



Let $A(a, b)$ be any point on the parabola and point $P(h, k)$ divides FA in ratio $1 : 2$

$$\Rightarrow h = \frac{a+6}{3} \text{ and } k = \frac{b}{3}$$

$$\Rightarrow a = 3h - 6 \text{ and } b = 3k$$

$A(a, b)$ lies on parabola, so $b^2 = 12a$

$$9k^2 = 12(3h - 6)$$

$$k^2 = 4(h - 2)$$

Hence, locus of P is $y^2 = 4(x - 2)$.

Question11

Equation of the line touching both parabolas $y^2 = 4x$ and $x^2 = -32y$ is

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Options:

A. $x + 2y + 4 = 0$

B. $2x + y - 4 = 0$

C. $x - 2y - 4 = 0$

D. $x - 2y + 4 = 0$

Answer: D

Solution:

To find the equation of the line that touches both parabolas $y^2 = 4x$ and $x^2 = -32y$, we start by considering the line in the form $y = mx + c$.

For the parabola $y^2 = 4x$, the condition for the line to be tangent is:

$$c = \frac{1}{m}$$

Thus, the line can be expressed as:

$$y = mx + \frac{1}{m}$$

For the parabola $x^2 = -32y$, the condition for tangency is:

$$c = 8m^2$$

Therefore, the line in this case is:

$$y = mx + 8m^2$$

By equating the expressions for c , we have:

$$\frac{1}{m} = 8m^2$$

Solving this equation for m , we get:

$$m^3 = \frac{1}{8}$$

$$m = \frac{1}{2}$$

Substituting $m = \frac{1}{2}$ back into the line equation, we obtain:

$$y = \frac{x}{2} + 2$$

Rewriting this equation in a standard form, we have:

$$x - 2y + 4 = 0$$

Thus, the equation of the tangent line is:

$$x - 2y + 4 = 0$$

Question12

If the normal chord drawn at $(2a, 2a\sqrt{2})$ on the parabola $y^2 = 4ax$ subtends an angle θ at its vertex, then $\theta =$

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Options:

A. 45°

B. 90°

C. 135°

D. 60°

Answer: B

Solution:

The normal to the parabola $y^2 = 4ax$ at the point (x_1, y_1) can be expressed as:

$$y = \frac{-y_1}{2a}(x - x_1) + y_1$$

For the point $(2a, 2a\sqrt{2})$, the equation becomes:

$$\begin{aligned} y &= \frac{-2a\sqrt{2}}{2a}(x - 2a) + 2a\sqrt{2} \\ &= -\sqrt{2}(x - 2a) + 2a\sqrt{2} \\ y &= -\sqrt{2}x + 4a\sqrt{2} \end{aligned}$$

Substituting this expression into $y^2 = 4ax$, we get:

$$\begin{aligned} (-\sqrt{2}x + 4a\sqrt{2})^2 &= 4ax \\ 2(x - 4a)^2 &= 4ax \\ x^2 + 16a^2 - 8ax &= 2ax \\ x^2 + 16a^2 - 10ax &= 0 \end{aligned}$$

This factors as:

$$\begin{aligned} x^2 - 8ax - 2ax + 16a^2 &= 0 \\ (x - 8a)(x - 2a) &= 0 \end{aligned}$$

Thus, $x = 8a$ or $x = 2a$.

Now, for $x = 8a$:

$$\begin{aligned} y &= -\sqrt{2} \times 8a + 4a\sqrt{2} \\ &= -4a\sqrt{2} \end{aligned}$$

For $x = 2a$:

$$\begin{aligned}
 y &= -\sqrt{2} \cdot 2a + 4a\sqrt{2} \\
 &= 2a\sqrt{2}
 \end{aligned}$$

The chord lies between points $(2a, 2a\sqrt{2})$ and $(8a, -4a\sqrt{2})$.

The slope m of this chord is:

$$m = \frac{-4a\sqrt{2} - 2a\sqrt{2}}{8a - 2a} = -\sqrt{2}$$

Considering the line from the vertex $(0, 0)$ to $(2a, 2a\sqrt{2})$, the slope is $\sqrt{2}$.

The angle θ between two lines with slopes m_1 and m_2 is calculated by:

$$\begin{aligned}
 \tan \theta &= \frac{m_1 - m_2}{1 + m_1 \cdot m_2} \\
 &= \frac{\sqrt{2} + \sqrt{2}}{1 - \frac{2}{2}} \\
 &= \frac{2\sqrt{2}}{0}
 \end{aligned}$$

Since $\tan \theta = \tan \frac{\pi}{2}$, this implies:

$$\theta = \frac{\pi}{2} = 90^\circ$$

Question13

If the ordinates of points P and Q on the parabola $y^2 = 12x$ are in the ratio $1 : 2$. Then, the locus of the point of intersection of the normals to the parabola at P and Q is

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Options:

A. $y + 18\left(\frac{x-6}{21}\right)^{\frac{3}{2}} = 0$

B. $y - 18\left(\frac{x-6}{12}\right)^{\frac{3}{2}} = 0$

C. $y + 12\left(\frac{x-6}{14}\right)^{\frac{1}{2}} = 0$

D. $y - 12\left(\frac{x-6}{18}\right)^{\frac{1}{2}} = 0$

Answer: A

Solution:

Given the equation of the parabola:

$$y^2 = 12x$$

By comparing this with the standard form $y^2 = 4ax$, we find:

$$a = 3$$

Consider the coordinates of the points P and Q on the parabola:

$$P = (at_1^2, 2at_1) = (3t_1^2, 6t_1)$$

$$Q = (at_2^2, 2at_2) = (3t_2^2, 6t_2)$$

Given that the ordinates of P and Q are in the ratio 1:2, we have:

$$\frac{6t_2}{6t_1} = 2$$

This simplifies to:

$$t_2 = 2t_1$$

Let $R(x, y)$ be the point of intersection of the normals at P and Q .

For a point t on the parabola $y^2 = 4ax$, the point of intersection of the normals at t_1 and t_2 is given by:

$$(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))$$

Substituting the values, we have:

$$x = 3(t_1^2 + t_2^2 + t_1t_2 + 2)$$

Substitute $t_2 = 2t_1$:

$$x = 3(t_1^2 + (2t_1)^2 + t_1 \cdot 2t_1 + 2)$$

$$x = 3(t_1^2 + 4t_1^2 + 2t_1^2 + 2) = 3(7t_1^2 + 2)$$

$$x = 21t_1^2 + 6$$

From this, we can express:

$$t_1^2 = \frac{x-6}{21}$$

Now, calculate y :

$$y = -3t_1 \cdot 2t_1(3t_1)$$

$$y = -18t_1^3$$

Substituting $t_1^2 = \frac{x-6}{21}$:

$$t_1 = \left(\frac{x-6}{21}\right)^{1/2}$$

Therefore:

$$y = -18\left(\frac{x-6}{21}\right)^{3/2}$$

Hence, the locus of the point of intersection of the normals at P and Q is:

$$y + 18\left(\frac{x-6}{21}\right)^{3/2} = 0$$

Question14

A common tangent to the circle $x^2 + y^2 = 9$ and parabola $y^2 = 8x$ is

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Options:

A. $3x - \sqrt{3}y + 2 = 0$

B. $x - \sqrt{3}y + 6 = 0$

C. $2x - \sqrt{3}y + 3 = 0$

D. $x - 3y + 6 = 0$

Answer: B

Solution:

To find the common tangent to the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$, we start with the equation of the tangent to the parabola. The general form for the tangent to $y^2 = 8x$ is given by:

$$y = mx + \frac{2}{m}$$

This tangent must also touch the circle $x^2 + y^2 = 9$. The condition for this line to be tangent to the circle is:

$$\left| \frac{2}{m\sqrt{1+m^2}} \right| = 3$$

Simplifying the equation, we get:

$$\frac{4}{m^2(1+m^2)} = 9$$

Solving this, we arrive at the polynomial equation:

$$9m^4 + 9m^2 - 4 = 0$$

By factoring, we find:

$$9m^4 + 12m^2 - 3m^2 - 4 = 0$$

This can be factored further into:

$$(3m^2 + 4)(3m^2 - 1) = 0$$

We obtain:

$$m^2 = \frac{1}{3}$$

Note that the solution $3m^2 + 4 = 0$ is rejected since it doesn't provide a valid, real number for m^2 .

Thus, the tangent can be expressed as:

$$y = \pm \frac{1}{\sqrt{3}}x \pm 2\sqrt{3}$$

This translates to:

$$\sqrt{3}y = \pm(x + 6)$$

The resulting equations are:

$$x \pm \sqrt{3}y + 6 = 0$$

Thus, the common tangents are:

$$x + \sqrt{3}y + 6 = 0 \quad \text{or} \quad x - \sqrt{3}y + 6 = 0$$

Question15

The normal drawn at a point $(2, -4)$ on the parabola $y^2 \pm 8x$ cuts again the same parabola at (α, β) , then $\alpha + \beta =$

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Options:

- A. 8
- B. 16
- C. 24
- D. 30

Answer: D

Solution:

To find where the normal at the point $(2, -4)$ on the parabola $y^2 = 8x$ intersects the same parabola again, we proceed as follows:

Equation of the Normal:

The equation of the normal to the parabola $y^2 = 8x$ at a point (x_1, y_1) is given by $y - y_1 = -\frac{2x_1}{y_1}(x - x_1)$.

Here, the point is $(2, -4)$, so $x_1 = 2$ and $y_1 = -4$.

Substituting into the normal equation:

$$y + 4 = -\frac{4}{-4}(x - 2)$$

$$y + 4 = x - 2$$

$$x - y = 6$$

Thus, $x = y + 6$.

Substitute Back into the Parabola's Equation:

Substituting $x = y + 6$ into the parabola equation $y^2 = 8x$:

$$y^2 = 8(y + 6)$$

$$y^2 = 8y + 48$$

$$y^2 - 8y - 48 = 0$$

Solve the Quadratic Equation:

Factor the quadratic:

$$(y - 12)(y + 4) = 0$$

The solutions are $y = 12$ and $y = -4$.

Find the Corresponding x Values:

For $y = 12$, $x = 12 + 6 = 18$.

We already know for $y = -4$, $x = 2$, which is the original point.

Conclusion:

The other intersection point is $(\alpha, \beta) = (18, 12)$.

Thus, $\alpha + \beta = 18 + 12 = 30$.

Question 16

If the axes are rotated through an angle 45° about the origin in anticlockwise direction, then the transformed equation of $y^2 = 4ax$ is

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Options:

A. $(x + y)^2 = 4\sqrt{2}a(x - y)$

B. $(x - y)^2 = 4\sqrt{2}a(x + y)$

C. $(x - y)^2 = \frac{43}{\sqrt{2}}(x - y)$

D. $(x + y)^2 = \frac{4a}{\sqrt{2}}(x - y)$

Answer: A

Solution:

To solve the problem of transforming the equation $y^2 = 4ax$ when the coordinate axes are rotated by 45° anticlockwise about the origin, we start by expressing the new coordinates (x, y) in terms of the original coordinates (X, Y) .

Given that the coordinates are rotated 45° anticlockwise, we can use the following transformation equations:

$$x = X \cos 45^\circ - Y \sin 45^\circ = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}} = \frac{X-Y}{\sqrt{2}}$$

$$y = X \sin 45^\circ + Y \cos 45^\circ = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}} = \frac{X+Y}{\sqrt{2}}$$

Substitute these expressions into the given equation $y^2 = 4ax$:

$$\left(\frac{X+Y}{\sqrt{2}}\right)^2 = 4a\left(\frac{X-Y}{\sqrt{2}}\right)$$

Simplifying the equation, we get:

$$\frac{(X+Y)^2}{2} = 4a\frac{(X-Y)}{\sqrt{2}}$$

Multiply through by 2 to eliminate the denominator:

$$(X + Y)^2 = 4\sqrt{2}a(X - Y)$$

Thus, the transformed equation is:

$$(x + y)^2 = 4\sqrt{2}a(x - y)$$

This is the equation of the curve after rotating the axes by 45° anticlockwise.

Question17

The line $x - 2y - 3 = 0$ cuts the parabola $y^2 = 4ax$ at the points P and Q . If the focus of this parabola is $(\frac{1}{4}, k)$. then $PQ =$

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Options:

A. $16a\sqrt{5}$

B. $8a\sqrt{5}$

C. $4a\sqrt{5}$

D. $2a\sqrt{5}$

Answer: A

Solution:

Focus of the parabola $y^2 = 42x$ is $(\alpha, 0)$

Here, $a = \frac{1}{4}$ and $K = 0$

So, equation of parabola is $y^2 = x$

As, the line $x = 2y + 3$ cuts the parabola

$$y^2 = x$$

$$y^2 = 2y + 3$$

$$\Rightarrow y^2 - 2y - 3 = 0$$

$$y^2 - 3y + y - 3 = 0$$

$$y = 3, -1$$

If $y = 3$, then $x = 9$ and

If $y = -1$, then $x = 1$

The two intersecting points

P and Q are $(9, 3)$ and $(1, -1)$

$$PQ = \sqrt{(9-1)^2 + (3+1)^2}$$

$$= \sqrt{64 + 16} = \sqrt{80}$$

$$= 4\sqrt{5}$$

$$= 16a\sqrt{5} \quad \left[\because a = \frac{1}{4} \right]$$

Question18

Which of the following represents a parabola?

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Options:

A. $x = 4 \cos t, y = 4 \sin t$

B. $x^2 - 2 = -2 \cos t, y = \cos^2 \left(\frac{t}{2} \right)$

C. $\sqrt{x} = \tan t, \sqrt{y} = \sec t$

D. $x = \sqrt{1 - \sin t}, y = \sin \left(\frac{t}{2} \right) + \cos \left(\frac{t}{2} \right)$

Answer: B

Solution:

From option (a).

$$x = 4 \cos t, y = 4 \sin t$$

$$x^2 = 16 \cos^2 t, y^2 = 16 \sin^2 t$$

So, $x^2 + y^2 = 16$, so this equation of a circle.

From option (b),

$$x^2 - 2 = -2 \cos t, y = \cos^2 \left(\frac{t}{2} \right)$$

$$x^2 = 2 - 2 \cos t \quad \dots \text{(i)}$$

$$y = \frac{1 + \cos t}{2} \quad \dots \text{(ii)}$$

From Eq. (ii),

$$\cos t = 2y - 1$$

$$\text{Now, } x^2 = 2 - 2 \cos t$$

$$x^2 = 2 - 2(2y - 1) \Rightarrow x^2 = 2 - 4y + 2$$

$$x^2 = 4 - 4y$$

So, this is represent the parabola.

Question19

Suppose a parabola passes through $(0, 4)$, $(1, 9)$ and $(4, 5)$ and has its axis parallel to the Y -axis. Then, the equation of the parabola is

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Options:

A. $19x^2 + 12y - 79x - 48 = 0$

B. $19x^2 + 12y - 79x + 48 = 0$

C. $19y^2 + 12x - 79y - 48 = 0$

D. $19y^2 + 12x - 79y + 48 = 0$

Answer: A

Solution:

Assume the vertex of the parabola is (h, k) and the length of its latus rectum is $4a$. Since its axis is parallel to the Y -axis, the equation of the parabola can be written as:

$$(x - h)^2 = 4a(y - k) \quad \dots (i)$$

The parabola passes through the points $(0, 4)$, $(1, 9)$, and $(4, 5)$. Substituting these points into equation (i), we get:

$$\therefore (0 - h)^2 = 4a(4 - k)$$

$$\Rightarrow h^2 = 4a(4 - k) \quad \dots (ii)$$

$$(1 - h)^2 = 4a(9 - k)$$

$$\Rightarrow 1 - 2h + h^2 = 4a(9 - k) \quad \dots (iii)$$

$$(4 - h)^2 = 4a(5 - k)$$

$$\Rightarrow 16 - 8h + h^2 = 4a(5 - k) \quad \dots (iv)$$

Next, we subtract Eqs. (ii) and (iii), and Eqs. (iii) and (iv), giving us:

$$1 - 2h = 20a \quad \dots (v)$$

$$15 - 6h = -16a \quad \dots (vi)$$

By solving Eqs. (v) and (vi), we obtain $a = -\frac{3}{19}$ and $h = \frac{79}{38}$.

Substituting these values back, we find $k = \frac{9889}{912}$.

Therefore, the equation of the parabola becomes:

$$\left(x - \frac{79}{38}\right)^2 = \frac{-12}{19}\left(y - \frac{9889}{912}\right)$$

$$\Rightarrow 19x^2 + 12y - 79x - 48 = 0$$

Question20

Suppose a parabola with focus at $(0, 0)$ has $x - y + 1 = 0$ as its tangent at the vertex. Then, the equation of its directrix is

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Options:

A. $x - y + 2 = 0$

B. $x - y - 2 = 0$

C. $x - y + 3 = 0$

D. $x - y + 4 = 0$

Answer: A

Solution:

Focus $\equiv (0, 0)$

Tangent at vertex of parabola : $x - y + 1 = 0$

The distance between the focus and tangent at the

$$\text{vertex} = \left| \frac{0-0+1}{\sqrt{1^2+1^2}} \right| = \frac{1}{\sqrt{2}}$$

The directrix is the line parallel to the tangent at vertex and at a distance $2 \times \frac{1}{\sqrt{2}}$ from focus.

Let the equation of directrix be

$$x - y + \lambda = 0$$

$$\text{Here, } \frac{\lambda}{\sqrt{1^2+1^2}} = \frac{2}{\sqrt{2}}$$

$$\Rightarrow \lambda = 2$$

$\therefore$ Equation of directrix is $x - y + 2 = 0$

Question21

If $ax + by = 1$ is a normal to the parabola $y^2 = 4px$, then the condition is

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Options:

A. $4ab = a^2 + b^2$

B. $4pab + ab^3 = a^2b^2$

C. $pa^3 = b^2 - 2pab^2$

D. $pa^2 + 4pa = a + b$

Answer: C

Solution:

Parabola : $y^2 = 4px$... (i)

Normal : $ax + by = 1$... (ii)

$$\Rightarrow y = -\frac{a}{b}x + \frac{1}{b}$$

Slope, $m = -\frac{a}{b}$ and constant, $c = \frac{1}{b}$

Condition of normal says that

$$c = -2m \left(\frac{\text{Coefficient of } x}{4} \right) - \left(\frac{\text{Coefficient of } x}{4} \right) m^3$$

$$\Rightarrow \frac{1}{b} = -2 \left(\frac{-a}{b} \right) \left(\frac{4p}{4} \right) - \left(\frac{4p}{4} \right) \left(\frac{-a}{b} \right)^3$$

$$\Rightarrow \frac{1}{b} = \frac{2pa}{b} + \frac{pa^3}{b^3}$$

$$\Rightarrow b^2 = 2pab^2 + pa^3$$

$$\Rightarrow pa^3 = b^2 - 2pab^2$$

Question22

The point of intersection of the latus rectum and axis of the parabola $y^2 + 4x + 2y - 8 = 0$ is

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Options:

A. $(\frac{9}{4}, -1)$

B. $(\frac{5}{4}, -1)$

C. $(\frac{7}{2}, \frac{5}{2})$

D. $(\frac{-5}{4}, 1)$

Answer: B

Solution:

Given parabola,

$$y^2 + 4x + 2y - 8 = 0 \dots (i)$$

Point of intersection of latusrectum and axis of parabola is foci.

Now, from Eq. (i), we get

$$(y + 1)^2 - 1 + 4x - 8 = 0$$

$$\Rightarrow (y + 1)^2 = -4x + 9$$

$$\Rightarrow (y + 1)^2 = -4(x - \frac{9}{4})$$

Compare with general standard form of parabola

$$(y - k)^2 = -4a(x - h)$$

where, foci = $(-a, 0)$

Here, $4a = +4$

$$\Rightarrow a = +1$$

$$\text{Then, } (x - \frac{9}{4}) = -1 \Rightarrow x = -1 + \frac{9}{4} = \frac{5}{4}$$

$$\text{and } (y + 1) = 0 \Rightarrow y = -1$$

$$\therefore \text{Focus} = (\frac{5}{4}, -1) = \text{Point of intersection}$$

Question23

The coordinates of the focus of the parabola described parametrically by $x = 5t^2 + 2$ and $y = 10t + 4$ (where t is a parameter) are

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Options:

A. (7, 4)

B. (3, 4)

C. (3, -4)

D. (-7, 4)

Answer: A

Solution:

$$x = 5t^2 + 2 \dots (i)$$

$$y = 10t + 4 \Rightarrow t = \frac{y-4}{10}$$

$$\text{From Eq. (i), } (x - 2) = 5 \left(\frac{y-4}{10} \right)^2$$

$$\Rightarrow (y - 4)^2 = 20(x - 2) \quad \begin{cases} x - 2 = 5 \\ y - 4 = 0 \end{cases}$$

$$\Rightarrow (7, 4) \text{ focus}$$

Question24

Find the equation of the parabola which passes through (6, -2), has its vertex at the origin and its axis along the Y-axis.

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Options:

A. $y^2 = 18x$

B. $x^2 = 18y$

C. $y^2 = -18x$

D. $x^2 = -18y$

Answer: D

Solution:

Equation of parabola whose vertex is at origin and axis is Y -axis is

$$x^2 = 4ay$$

This equation passes through $(6, -2)$

$$\begin{aligned}\therefore 6^2 &= 4a(-2) \\ \Rightarrow \frac{-36}{8} &= a \Rightarrow a = \frac{-9}{2}\end{aligned}$$

$\therefore$ Required parabola $x^2 = -18y$

Question25

If one end of focal chord of the parabola $y^2 = 8x$ is $(\frac{1}{2}, 2)$, then the length of the focal chord is units.

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Options:

A. $\frac{625}{4}$

B. $\frac{5}{\sqrt{2}}$

C. $\frac{25}{2}$

D. 25

Answer: C

Solution:

Given parabola, $y^2 = 8x$

Comparing with $y^2 = 4ax$

We have, $a = 2$

One end of focal chord $\left(\frac{1}{2}, 2\right)$

Parametric form of ends of focal chord $= (at_1^2, 2at)$

$$\Rightarrow 2at = 2$$

$$\therefore 2 \cdot 2 \cdot t = 2 \Rightarrow t = \frac{1}{2}$$

$$\begin{aligned}\text{Length of focal chord} &= a \left(t + \frac{1}{t} \right)^2 \\ &= 2 \left(\frac{1}{2} + 2 \right)^2 = 2 \cdot \left(\frac{5}{2} \right)^2 = \frac{25}{2}\end{aligned}$$
